

Estimated net acid excretion inversely correlates with urine pH in vegans, lacto-ovo vegetarians, and omnivores.



OBJECTIVE: Diet affects urine pH and acid-base balance. Both excess acid/alkaline ash (EAA) and estimated net acid excretion (NAE) calculations have been used to estimate the effects of diet on urine pH. This study's goal was to determine if free-living vegans, lacto-ovo vegetarians, and omnivores have increasingly acidic urine, and to assess the ability of EAA and estimated NAE calculations to predict urine pH. DESIGN: This study used a cross-sectional design. SETTING AND PARTICIPANTS: This study assessed urine samples of 10 vegan, 16 lacto-ovo vegetarian, and 16 healthy omnivorous women in the Boston metropolitan area. Six 3-day food records from each dietary group were analyzed for EAA content and estimated NAE, and correlations with measured urine pH were calculated. RESULTS: The mean ($\pm$ SD) urine pH was 6.15 $\pm$ 0.40 for vegans, 5.90 $\pm$ 0.36 for lacto-ovo vegetarians, and 5.74 $\pm$ 0.21 for omnivores (analysis of variance, $P = .013$). Calculated EAA values were not significantly different among the three groups, whereas mean estimated NAE values were significantly different: 17.3 $\pm$ 14.5 mEq/day for vegans, 31.3 $\pm$ 8.5 mEq/day for lacto-ovo vegetarians, and 42.6 $\pm$ 13.2 mEq/day for omnivores (analysis of variance, $P = .01$). The average deattenuated correlation between urine pH and EAA was 0.333; this value was -0.768 for estimated NAE and urine pH, with a regression equation of $\text{pH} = 6.33 - 0.014 \text{ NAE}$ ($P = .02$, $r = -0.54$). CONCLUSIONS: Habitual diet and estimated NAE calculations indicate the probable ranking of urine pH by dietary groups, and may be used to determine the likely acid-base status of an individual; EAA calculations were not predictive of urine pH.